



GROW_SCI_A2_W02 · GROW Science teacher note

Staff-facing; review only. Verbatim SoW objective: Describe the Sun, Earth and Moon as approximately spherical bodies.

Source: GROW scheme of work, Autumn 2 week 2. Pupil goal: show how a ball models the Sun, Earth and Moon.

40-minute sequence and success

Arrival 5 · Starter 3 · I do 5 · We do 7 · We do 2 5 · Independent 8 · Review 4 · Exit 3 = 40.

One-minute resets are inside We do 2, Review and Exit. Success: I can choose a ball to model a round body., I can turn it to show more than one side., I can name one limit of a model.

Retrieval and I-do check

When	Question	Answer	SoW cell
last lesson	What makes day turn to night at one place?	Earth turns on its axis.	GROW scheme of work, Autumn 2 week 1
2-4 weeks	What goes around the Sun?	Earth and other planets.	GROW scheme of work, Autumn 1 week 6
last term	Name one force you have explored.	Gravity, air resistance or friction.	GROW scheme of work, Autumn 1 week 1

Autumn 1 is the prior half-term. Confirm actual taught coverage before using a retrieval question. Show me: Point: which object can I turn to see another curved side? Answer: The ball. All pupils show by point, card or board.

Words (same as page and organiser):

Word	Tier	Simple definition	Example sentence
spherical	Tier 3	shaped roughly like a ball	Earth is approximately spherical.
curve	Tier 2	a line or surface that bends	The ball has a curved surface.
shadow	Tier 2	a dark area where an object blocks light	The ball makes a shadow on the card.
model	Tier 3	a simple way to show an object or idea	A ball is a model of Earth's shape.

Faded example, tiered task and stretch

I do full: "Ball → turn it → see another curved side → useful for an approximately spherical body; ball sizes and gaps are not to scale." We do part-done: "Ball → turn it → see → useful because ."

Removed step: Name the new curved side and explain why it helps. Independent: no filled step. Next fade: Next time remove the ball cue and ask for a model choice before showing kit.

We do F: Point to the ball and the paper edge before you turn the model. M: Turn the ball and explain what new view it gives. H: Explain why one photo can look circular but cannot show the full solid shape. Independent F: Choose a ball rather than a flat circle. M: Turn the ball and explain what new view it gives. H: Explain what a ball model reveals and how its scale may mislead. Go deeper:



Compare a ball and a photo. What can each show, and what does each hide? Named harder stretch — Model limits: A photo shows Earth as a circle. Choose another observation or model to test its full shape. Explain why one photo is not enough.

The model, source or practical route is: safe balls, paper circles, torch. Two on-screen interactions are card sort and What can show another curved side when turned?; native keyboard controls, live feedback, no storage. The diagram is original and simplified, not to scale. Its labels are checked in the facts ledger.

Practical: Predict first: Predict which object can turn to show a different curved side: ball or paper circle? Kit: Three safe balls or modelling spheres, Paper circles, Torch with adult-controlled beam, Tray or stable surface. Steps: Compare a ball with a flat circle; Turn the ball once and record what new side you see; Turn the paper circle and record what it shows; Put the torch on the ball; do not point it into eyes. Results table: Shape model | new curved side seen?, Ball | yes/no, Paper circle | yes/no. Check against the school risk assessment and CLEAPSS guidance. Keep the torch beam out of eyes; use safe, stable kit.

Hinge and decision

Question: A class models the Moon's whole shape. Which object is the closest simple shape?

Option	Correct?	Diagnosis
A thin flat disc.	no	Uses a flat circle for a three-dimensional Moon.
A long straight stick.	no	Mistakes a long one-dimensional shape for a round body.
A round solid ball.	yes	Uses the new case accurately.
A flat square sheet.	no	Chooses a flat square because the diagram is on paper.

Rule: About 3 in 4 or more right → move on. Otherwise re-model with the named I-do example, then ask the parallel hinge. Parallel: Which simple shape could model the Sun's whole shape: ball or flat ring? Answer: ball; neither size nor gaps are to scale.

Misconceptions:

Misconception	Spot	Correct
A circle on paper is a sphere.	Pupil chooses A, "A thin flat disc."	Turn a ball to reveal a new side; a flat circle cannot do that.
Spherical means perfectly smooth and exactly round.	Pupil describes C as perfectly smooth and exactly round.	Say approximately spherical; the real bodies are not exact toy balls.

AQA Entry Level Science style bridge (classroom planning): Describe what you see when you turn a shape model. This is a planning bridge, not a claimed unit outcome or code.

Feedback codes: VF · WS · I · NS(+) · E · R · // · ?. At We do 2, record WS or I and actual prompt level; at Review, VF and specific NS(+); at Exit, one self-edit // or read back ?. R means a response to the next step heard by an adult in the next lesson, never auto-awarded. E: evidence captured on the school's evidence app only if an observable curriculum action occurred, never while distressed. The page stores no response. Pathways: high support with VF/WS; Build: mix of WS/I; Grow: mostly I and self-edit; Launch: independent self-assessment.

Planned adaptations



Broad area	EEF five-a-day strategy	Specific support
Cognition & Learning	scaffolding	Give a tactile ball and a two-step turn card, then remove the turn cue.
Communication & Interaction	cognitive/metacognitive strategies	Let pupils rehearse the M or H model explanation with a partner and pointing cue.
SEMH	flexible grouping	Offer a safe kit role and quiet pause cue before the model comparison.
Sensory/Physical	technology	Use a large stable ball or textured ball; allow a photo of the pupil's real demonstration.
EAL (language, not SEN)	explicit instruction	Teach spherical with a ball and gesture. Keep M/H open with the speaking frame and home-language planning.

Keep M/H challenge open for EAL with the same words and home-language planning. Pupil may point, sign, speak, draw, type, write, use a scribe or have a staff-captured photo. Offer speech-to-text, TTS or immersive reader on the staff device.

Adults, regulation and prompt ladder

Wait and count to 5. → Point to the lesson's diagram, kit or source. → Ask: What is done, and what is the next step? → Model one step, then return the next step to the pupil. → Finish one example together, then fade the help. No physical prompt. Stage scripts: I do "Watch the ball, then point to the side you can see."; We do "What is done? What does turning show next?"; Independent "Show your model choice. What can you explain yourself?"

At Independent, one adult works with supported pupils, the other holds the group; choose roles at the start and swap across the week. If only one adult attends, use the actual staffing plan. At Exit, hand off the deepest prompt level and what was independent.

Cold-call / invite bank, answers:

- F: Point to the solid shape. → The ball.; Point to the flat shape. → The paper circle.
- M: What changes when you turn the ball? → The side we can see.; Why is the ball a useful shape model? → It has a curved surface on more than one side.
- H: What is one limit of the toy-ball model? → Its size, texture or distance may not match the real body.; Why is one flat photo not enough? → It only shows one view of a solid body.

Regulation: Agree a quiet pause signal and supervised place. Check in before hinge and Independent; point to the last completed step on re-entry. No evidence while a pupil is distressed.

lesson record: GROW_SCI_A2_W02 via staff device or projector only. Gatsby 4: Spacecraft designer: use shape models and state what they leave out.

Reading gate: GROW 8-11; rough visible-body ARI grade 2.83; age guide 8.83 years; rough FK grade 3.88; 838 pupil-visible words excluding marked adult guidance and navigation. Screening estimate only. This is a rough screen, not an individual assessment.

Pupil-facing risk: Torch beams can dazzle eyes; round models may roll off the table.

Pupil-facing safety: Keep the beam out of eyes. Turn models over the tray and put them back.

SLT planning and evaluation

Lesson code: SCI-GROW-A2-W02

SoW reference: GROW scheme of work, Autumn 2 week 2



Exact SoW outcome: Describe the Sun, Earth and Moon as approximately spherical bodies.

Evidence (scheme of work): Pearson Edexcel Entry Level Certificate in Science 8939 (E1→E3); AQA UAS science units. F: working-scientifically observation & results tables. S: end-of-unit assessment + Entry Level / UAS evidence.

Evidence in this lesson: Pearson Edexcel Entry Level Certificate in Science 8939 (E1→E3); AQA UAS science units. F: working-scientifically observation & results tables. S: end-of-unit assessment + Entry Level / UAS evidence.

Where it shows in books: Independent observation/result table or labelled explanation; review edits and exit response beside the original work.

Label evidence on the school's evidence app with this code: SCI-GROW-A2-W02. For spoken, practical or photographic evidence, record the same code with the evidence.

Arrival (5 min): Sort a coin, paper circle and ball: flat or solid? Staff: Offer a predictable seat or standing spot and the first visual cue. No public test.

Starter (3 min): Think back. Say, point or write one answer for each question. Staff: Ask three low-stakes retrieval questions; prior coverage must be checked.

I Do (5 min): Watch me turn a ball. I can see another curved side. A paper circle is flat. A ball helps show the shape of the Sun, Earth and Moon, but its size is not exact. Staff: Model the full reasoning step, pre-teach four words and ask the show-me question.

We Do (7 min): Use a ball and a paper circle. The model and first label are ready. Turn the ball and complete the missing label: new side or the same flat face? Staff: Show the part-done example, then allow paired talk before an answer. One-minute reset inside this stage.

We Do 2 (5 min): Show everyone which shape model lets us turn to see another side. Staff: All pupils answer the four-option hinge; use the decision rule, not a score.

Independent (8 min): Turn a shape model. Explain one way it helps and one limit. You can show, say or draw. Staff: One adult supports agreed pupils, the other holds the group; swap across the week. One-minute reset inside this stage.

Review (4 min): Hear a comment on your model. Change one label or spoken reason and say what changed. Staff: Give a specific verbal comment (VF) and invite one self-edit (//) or read-back (?).

Exit (3 min): Show which shape model helps. Say one way it helps and one thing it cannot show exactly. Staff: Hear the answer and name how it changes the next example or support. One-minute reset and handoff.

Planned adaptation — cognition learning: Give a tactile ball and a two-step turn card, then remove the turn cue.

Planned adaptation — communication interaction: Let pupils rehearse the M or H model explanation with a partner and pointing cue.

Planned adaptation — semh: Offer a safe kit role and quiet pause cue before the model comparison.

Planned adaptation — sensory physical: Use a large stable ball or textured ball; allow a photo of the pupil's real demonstration.

Planned adaptation — eal: Teach spherical with a ball and gesture. Keep M/H open with the speaking frame and home-language planning.

Planned adaptation — eef strategy: cognition learning: scaffolding; communication interaction: cognitive/metacognitive strategies; semh: flexible grouping; sensory physical: technology; eal: explicit instruction

Staff: confirm the planned adaptations actually used and record any changes below.



If behind — must teach core: Turn a shape model. Explain one way it helps and one limit. You can show, say or draw. Preserve the individual evidence required above.

If behind — shorten or merge: After a secure hinge, shorten repeated oral rehearsal. If needed, merge the exit reflection into the Arrival of Autumn 2 W03. Keep the core evidence and any safety teaching; record the change and coverage impact.

If behind — next lesson: Autumn 2 W03: Research and present findings about the Solar System (ICT).

If behind — dependency: Pass on today's checked understanding and unresolved hinge errors for the next starter. Do not replace incomplete individual work with a group product.

Staff evaluation — date taught:

Staff evaluation — delivered as planned: ☐ Yes ☐ Partly ☐ No

Staff evaluation — criteria met: ☐ All ☐ Most ☐ Some

Staff evaluation — adaptations made and why:

Staff evaluation — change from sow reason and coverage impact:

Staff evaluation — evidence banked on the school's evidence app:

Individual target opportunities

Cognition & learning: We do: Name the new curved side and explain why it helps. Record the pupil's own sequencing, attention or independence target, including the prompt level needed.

Communication & interaction: Check: respond to "A class models the Moon's whole shape. Which object is the closest simple shape?" in the pupil's agreed mode. Evidence an individual communication/turn-taking target without requiring speech.

SEMH: Arrival: Sort a coin, paper circle and ball: flat or solid? Use the agreed predictable start or pause/re-entry plan; evidence the pupil's own regulation or engagement target privately.

Sensory/physical: Independent: Turn a shape model. Explain one way it helps and one limit. You can show, say or draw. Offer the agreed accessible input, enlarged material or adapted equipment; record the pupil's target-related access/independence, not a diagnosis.

EAL: Model/shared rehearsal: use spherical, curve, shadow, model with the actual example; allow first-language planning and oral rehearsal before the independent response. Record the agreed language target; EAL is not SEN.

Capture E against the pupil's own IEDP target on the school's evidence app; the IEDP carries their EHCP-linked and Boxall targets, so one record serves every plan. Follow the pupil's IBSP where it differs from these routines.